Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 0 complete test ICs for all5 lambdas within ANN10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**0**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
1 ANN10	**1**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**10**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**100**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	**1000**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: . IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 0 complete test ICs for all5 lambdas within P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
2 P-only DeepONet10	**0**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**1**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**10**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**100**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**1000**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: . IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 0 complete test ICs for all5 lambdas within ANN10 + P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
3 ANN10 + P-only DeepONet10	**0**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
3 ANN10 + P-only DeepONet10	**1**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**10**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**100**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**1000**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: . IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 0 complete test ICs for all5 lambdas within T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
4 T/P DeepONet10	**0**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
4 T/P DeepONet10	**1**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	**10**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
4 T/P DeepONet10	**100**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	**1000**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: . IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 0 complete test ICs for all5 lambdas within ANN10 + T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
5 ANN10 + T/P DeepONet10	**0**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8/18
5 ANN10 + T/P DeepONet10	**1**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	**10**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	**100**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	**1000**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: . IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 0 complete test ICs for all5 lambdas within T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
6 T/P/ϕ DeepONet10	**0**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
6 T/P/ϕ DeepONet10	**1**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
6 T/P/ϕ DeepONet10	**10**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	**100**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	**1000**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: . IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 0 complete test ICs for all5 lambdas within ANN10 + T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
7 ANN10 + T/P/ϕ DeepONet10	**0**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12/18
7 ANN10 + T/P/ϕ DeepONet10	**1**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
7 ANN10 + T/P/ϕ DeepONet10	**10**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	**100**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	**1000**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: . IC sets differ between teams; use the all35-variant summary for architecture comparisons.